Smart AI Fan

A · Team & Institutional Context

TECHNOMANCERS	T-2026	Aditya Institute of Technology and Management, Tekkali	Electronics and Communication Engineering

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B · Problem & Idea Framing

Track 1 Fan — combining smart-home triggers, sensor-based multi-directional swing, and blade-design efficiency: one sensing-and-control system drives comfort automation and swing, while the blades set the operating point both act on.

Proposed idea/mechanism (one paragraph): a 1,400 mm-sweep ceiling fan. A local controller reads temperature/humidity to run comfort automation with manual-override precedence and defined safe fallback; a self-locking worm pan/tilt head steers the whole airflow disc (rotor stopped before tilt; full ±20° needs ≥ 2.660 m ceiling, level-only on 2.650–2.660 m); quasi-isotropic [0/±45/90]s carbon/epoxy blades run on a 24 V FOC BLDC drive, overseen by a locked-firmware supervisor MCU that can cut actuator power on any anomaly.

Baseline-vs-target, quantified: no measured baseline of ours exists. Versus a fixed-speed, fixed-direction, manual fan: steerable airflow (±20° tilt); automated response ≤ 5 s with ≥ 95 % success over 10–20 scenarios; a device-trigger path; defined safe fallback on sensor/network loss. Air delivery ≥ 245 m³/min and service value ≥ 4.1 m³/min/W are IS 374 targets; ≈ 8.7 m³/min/W is indicative (245 ÷ 28.3 W), not measured.

Other technical detail: safety scope in C; remaining risks in D.

C · Technical Feasibility

Sensors feed comfort automation and closed-loop FOC speed control (≤ 0.5 s). Plausibility gates (–10…+80 °C, ≤ 3 K/s) route implausible input to a safe default curve; a hardware NTC trips at 165 °C; the self-locking worm holds position on power loss, backed by encoders, limit switches and supervisor power-cut — three independent defenses. All subsystems are standard parts; a safe demonstrator is feasible now; rotor validation stays gated.

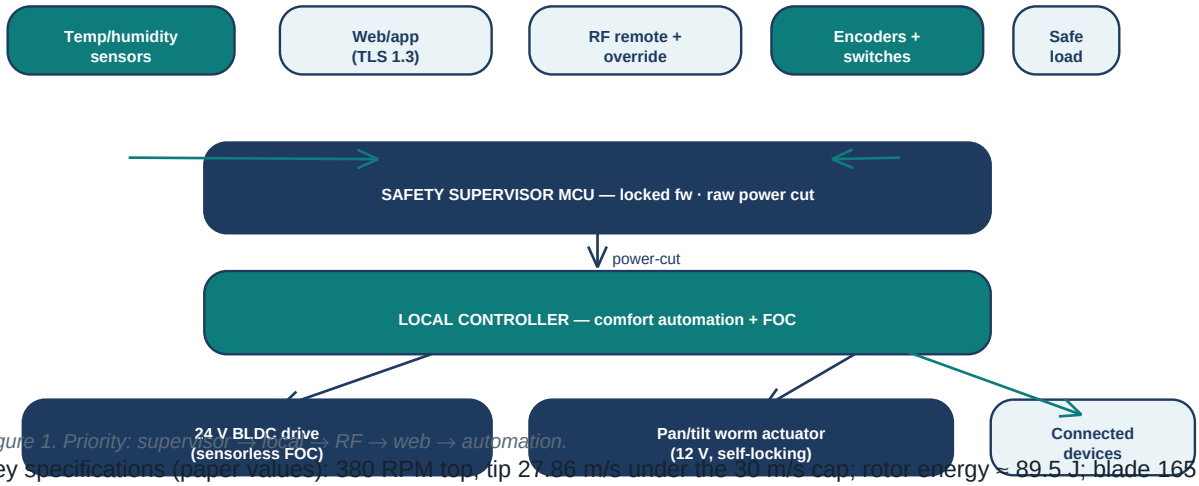

Figure 1. Priority: supervisor → local → RF → web → automation.

Key specifications (paper values): 380 RPM top, tip 27.86 m/s under the 30 m/s cap, rotor energy ≈ 89.5 J, blade 165 g; worm ≥ 6/2.5 N·m; anchor demand 139.9 N vs 1,500 N ultimate; cable 6.5 kN; electrical per IS 302-2-80 — compliance not established. 456 RPM is a hazard case, never a target.

D · Execution Feasibility

Rough components: composite rotor (custom), 24 V FOC BLDC motor + controller, supervisor + local MCU pair, temp/humidity sensors, worm actuators (custom-rated) with encoders/limit switches, 24 V SMPS, optional Wi-Fi/BLE, RF remote + override, device trigger or safe test load, instrumentation, safety cable.

Rough cost/BOM (order of magnitude, no quotations): ≈ INR 26,650 per fan; ≈ INR 5.8 lakh (3-fan programme). Sourcing risk: custom CFRP blades (tooling, cure, verification; no supplier datasheet yet); worm actuators need confirmation of self-lock at rated torque.

Build plan to Round 2: (1) low-voltage sensing + comfort-control demonstrator; (2) verify override precedence and sensor-disconnect/network-loss fallback; (3) device trigger on a safe test load; (4) record results against predefined acceptance tests; (5) motor/rotor integration for Round 2 only after safety review and approval.

Any other technical detail: paper design only — no hardware, nothing measured. No authorization for fabrication, energization, powered-rotor testing, installation, or certification.